

# The Blueprint for Smarter Advertising

*Driving Cost Efficiency in Digital Advertising*



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# Chapter 1: Overview & Problem Statement

# Audience

**Annie Chen**, Senior Marketing  
Manager, Apple Inc.



**John**, Data Analytics Manager,  
D Advertising Pte Ltd



# Problem Statement

- ▶ Apple is spending heavily in digital advertising for its products
- ▶ Want to increase its cost efficiency for digital advertising budget

# Do you know the difference?

- ▶ Increasing Cost Efficiency
- ▶ Minimizing Cost

# Goals

- ▶ Achieve at least 10% increase in expected online purchase of Apple products
- ▶ Achieve 20% decrease in cost per acquisition/purchase (CPA)
- ▶ Assume advertising budget unchanged

# Chapter 2: Data Sources & Methodology



# Data Source

- ▶ Fictitious digital advertising dataset (May - Aug 2025)
- ▶ Ad campaigns spend & event types from social media platforms, no revenue data
- ▶ Cleaned & removed duplicate records (approx. 1%)
- ▶ Created metrics: Cost of Acquisition (CPA), Conversion Rate (CR), Click-Through Rate (CTR)

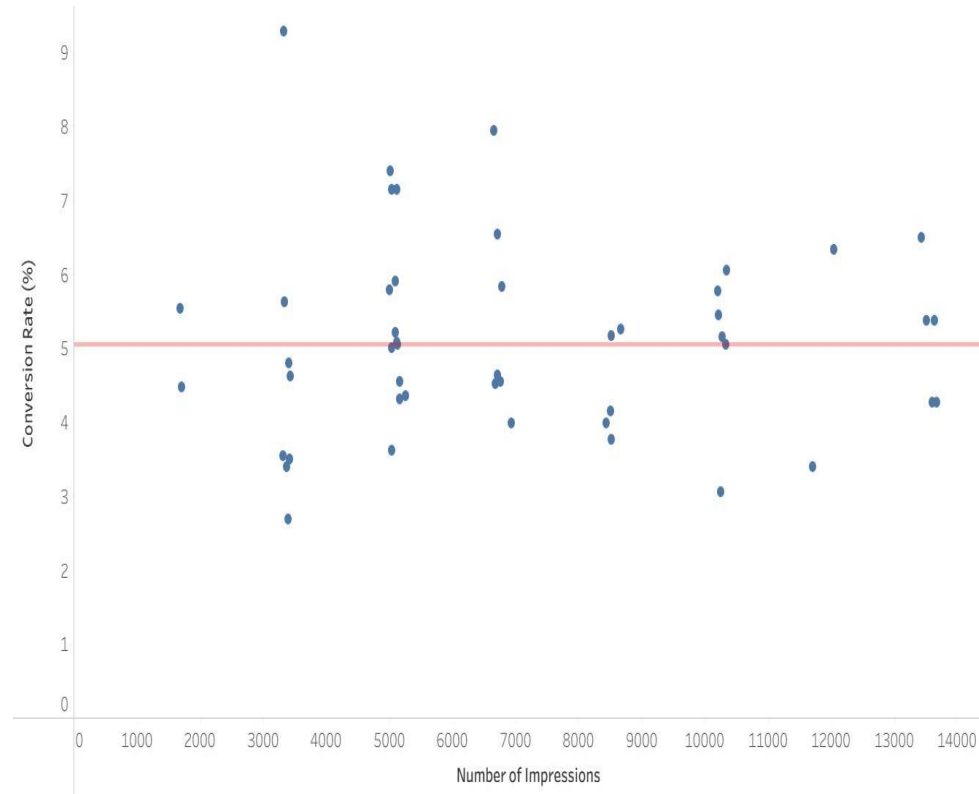
# Methodology

- ▶ Exploratory Data Analysis to identify trends, inefficiencies & outliers
- ▶ Predictive Modeling (Comparing Logistic Regression vs Light GBM) to predict purchase probabilities
- ▶ Tableau Visualization for Dashboard

# Chapter 3: Data Analysis & Insights

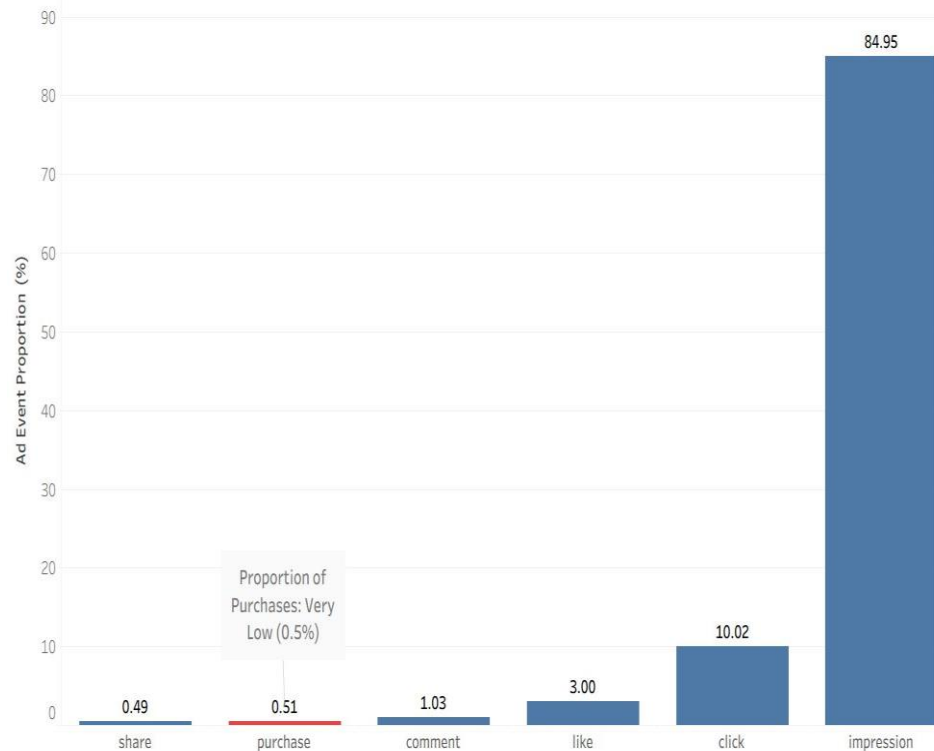
# Analysis: Conversion Rate & Ad Impressions

- Stable Ad Conversion Rate (3-6%) at all levels of impressions
- **Ad scaling is ineffective** as conversion rate doesn't increase with impressions
- Apple should **increase cost efficiency at current budget**, instead of increase ad budget



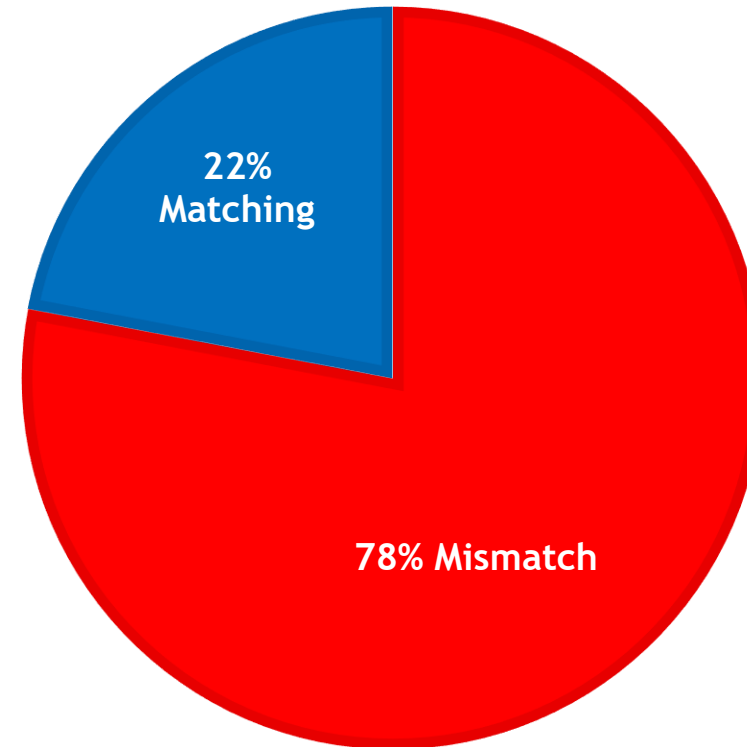
# Analysis: Low Purchase Rate

- Very low purchase rate (0.5%) as compared to industry of 2-4%
- Affect prediction modeling performance results
- In-depth research to find out cause & increase it



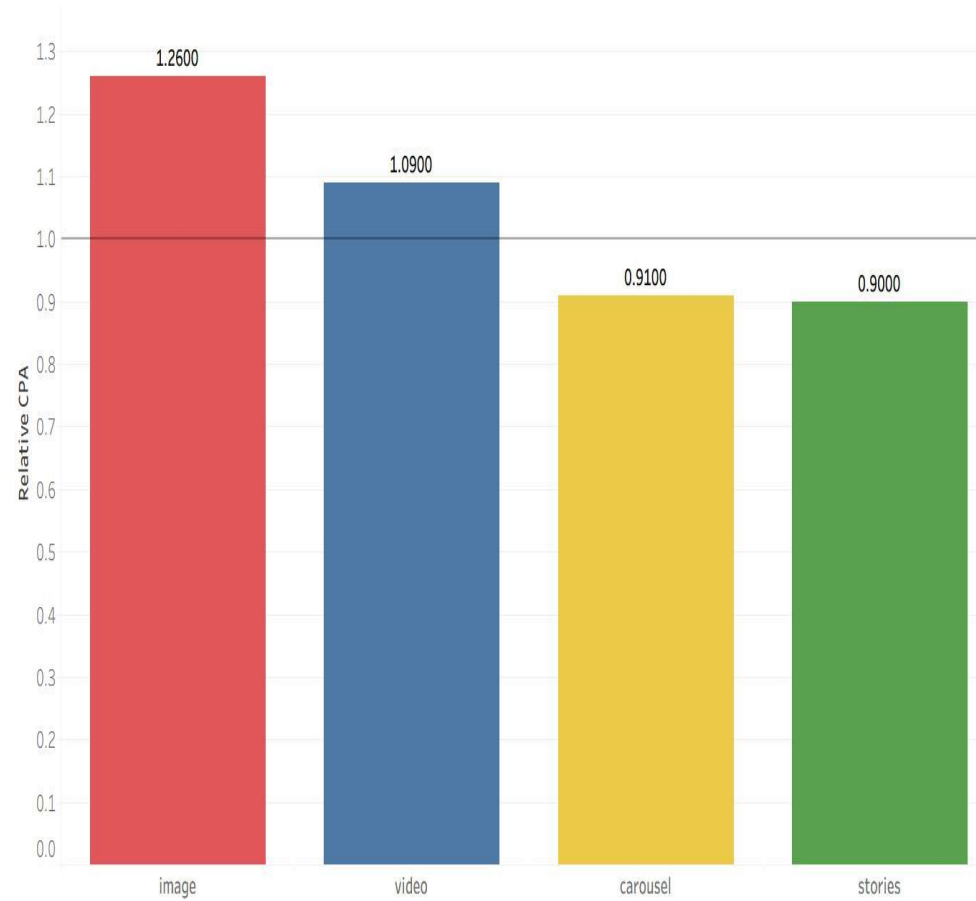
# Analysis: Target vs Actual Users' Age Group Segments

- High actual-target user age group mismatch at 78%
- Ineffective Age Group Targeting
- Improve on user data validation process to reduce mismatch



# Analysis: Relative CPA & Advertisement Type

- Image & Video have higher relative CPA compared to Carousel & Stories
- Ad Type Median CPA = 1
- Re-allocate budget from Image & Video to Carousel & Stories
- Cost Efficiency 



# Analysis: Countries & CTR, CR & CPA

- High Cost, Low Purchase  
Countries: Australia, Brazil
- Low Cost, High Purchase  
Countries: India
- High Cost, High Purchase  
Country: Japan
- Reduce allocation to Australia,  
Brazil, increase exposure to India

| Countries      | CPA    | CR     | CTR    | Outcomes |
|----------------|--------|--------|--------|----------|
| australia      | High   | Low    | Medium | Bad      |
| brazil         | High   | Low    | Low    | Bad      |
| canada         | Low    | Low    | High   | Good     |
| france         | Medium | High   | Low    | Bad      |
| germany        | Low    | Medium | High   | Good     |
| india          | Low    | High   | Low    | Bad      |
| japan          | High   | High   | Medium | Medium   |
| mexico         | Low    | Medium | High   | Good     |
| united kingdom | Medium | Low    | Low    | Bad      |
| united states  | Medium | Medium | Medium | Medium   |



# Correlation Analysis

- Very low correlations (0 - 0.01) among features (age group, country, days of week, ad types), & with Purchase Rate
- Due to low purchase rate (0.5%)
- No Multicollinearity issue

# Summary

**339K**

Impressions

**11.7%**

CTR

**5%**

Conversion Rate

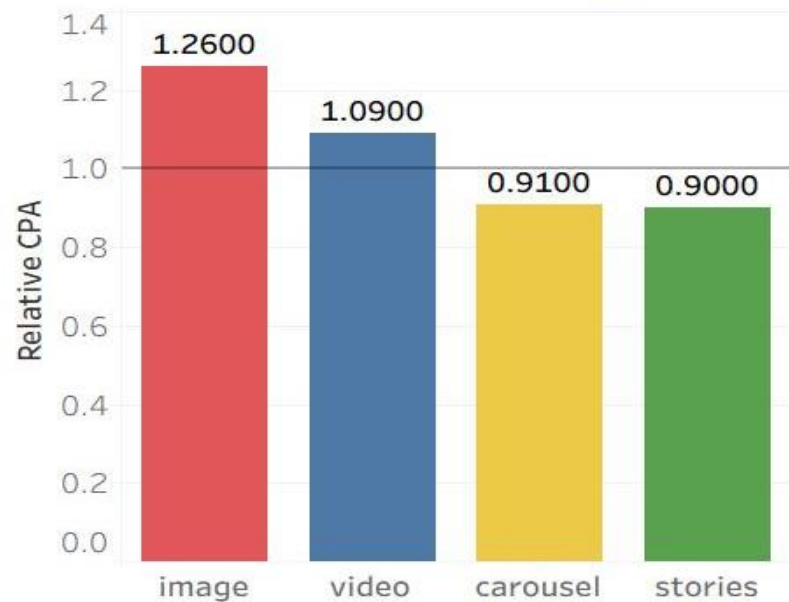
**0.5%**

Purchase Rate

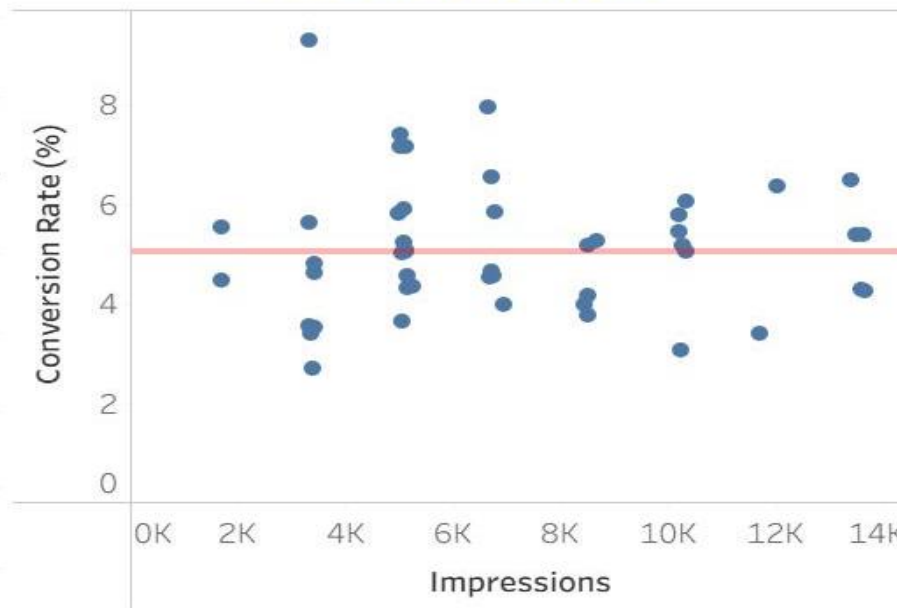
**\$958**

CPA per ad

**Relative CPA & Ad Type**



**CR & Impressions**



**Target Audience Mismatch**  
78% Off-Target



**Key Insight**  
Lower CPA for Stories Ad

# Chapter 4: Predictive Modeling & Demo

# Comparison of 2 Predictive Models

|                        | Logistic Regression (Log R)                                     | Light GBM (LG)                                                    | Winner                                                              |
|------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------|
| Ease of Interpretation | Very Easy<br>(Coefficients shows features' impact)              | Difficult<br>(features' importance is indirect)                   | Log R                                                               |
| Speed                  | Fast to train                                                   | Fast to train (requires hyperparameters tuning)                   | Tie                                                                 |
| Data Suitability       | Small to Medium Datasets<br>(<1m observations)                  | Large Datasets<br>(> 1m observations)                             | Log R                                                               |
| Performance            | Mid Precision (0.5)<br>Mid Recall (0.53)<br>Mid F1 Score (0.33) | Low Precision (0.01)<br>High Recall (0.96)<br>Low F1 Score (0.01) | Log R due to more balanced, lesser budget wastage (false positives) |

Predictive Model Chosen: **Logistic Regression**

# Predicted Results by Countries

- Countries & CPA have strongest impact on Purchase Probability
- Lowest Purchase & Highest CPA: France & Japan
- Highest Purchase & Lowest CPA: United States & United Kingdom

|                | expected_purchases | expected_cpa |
|----------------|--------------------|--------------|
| country        |                    |              |
| france         | 65.0               | 183.54       |
| japan          | 112.0              | 106.72       |
| mexico         | 113.0              | 104.48       |
| brazil         | 118.0              | 101.64       |
| australia      | 138.0              | 87.00        |
| germany        | 152.0              | 78.20        |
| india          | 183.0              | 64.85        |
| canada         | 194.0              | 60.98        |
| united kingdom | 283.0              | 42.18        |
| united states  | 617.0              | 19.35        |

# Demo on Predicting Cost of Acquisition (CPA)

CPA Prediction

# Chapter 5: Model Assumptions & Limitations

# Assumptions

- ▶ Online users' behavior patterns are representative of real world
- ▶ Market conditions remain unchanged
- ▶ Purchase variable is linearly related to features
- ▶ Features are independent of each other



# Limitations

- ▶ Low purchase rate (0.5%) ↓ model accuracy, predictions less reliable
- ▶ 4-month time period restricts visibility of seasonal or long-term digital ad trends
- ▶ Absence of ad revenue data restricts digital advertising analysis
- ▶ High mismatch of actual-target group segment hinders age group segment analysis

# Chapter 6: Recommendations & Next Steps

# Business Recommendations

- ▶ Re-allocate 20% digital ad spend from Image to Carousel & Stories ad types, result in 18%  in expected number of purchases
- ▶ Re-allocate their digital ad budget from high CPA cost countries (France) to low-cost countries (United States & India). CPA expected to  more than 50%

# Data & Model Improvements

- ▶ Ad revenue data should be included for comprehensive analysis
- ▶ Longer period data for seasonal trend analysis
- ▶ User target group data validation process should be improved
- ▶ In-depth tuning of Light GBM hyper-parameters

# Next Steps

- ▶ 20% of digital ad budget re-allocated from Video to Stories ads
- ▶ Pilot ad reallocation from France to US/India to test for 1 month
- ▶ User Target data validation process to be improved